

Dear Dr. Noor Amiri,

I am writing to express my strong interest in the fully funded PhD position in Robotics in the BRAINS Lab at the Department of Computer Science, The University of Alabama, beginning Spring 2027. I hold a Master's degree in Computer Science from a U.S. university and have a strong background in machine learning, deep learning, computer vision, artificial intelligence, and software development. I am particularly excited about the opportunity to contribute to research at the intersection of AI, robotics, computer vision, autonomous systems, and embodied intelligence.

My academic and professional experience has provided me with a strong foundation in developing and applying machine learning solutions to real-world problems. I have worked extensively with Python, machine learning algorithms, deep learning, data analysis, and software development, including developing AI-driven applications and integrating machine learning models into practical systems. Through these experiences, I have developed a strong interest in moving beyond purely software-based AI toward intelligent systems that can perceive, reason, learn, and interact with the physical world.

My research interests closely align with the BRAINS Lab's focus on robotic perception, computer vision, embodied AI, autonomous decision-making, multimodal perception, and intelligent robotic systems. I am particularly interested in developing computer vision and deep learning methods that allow robots to understand complex environments and make intelligent decisions. I am also interested in exploring multimodal learning, where visual, sensory, and other sources of information can be combined to improve robotic perception and interaction.

My previous work in machine learning and AI has given me experience with Python and deep learning methodologies, along with experience analyzing data, developing predictive models, performing exploratory data analysis, and evaluating model performance. I have also worked on research-oriented problems involving AI for cybersecurity and resource-constrained edge systems. These experiences strengthened my ability to formulate research questions, implement computational approaches, analyze experimental results, and communicate technical findings. I am eager to extend these skills into robotics, particularly through hands-on work with robotic arms, humanoid robots, ROS/ROS 2, and simulation environments.

What particularly attracts me to the BRAINS Lab is its emphasis on developing intelligent robotic systems and validating AI algorithms on real-world robotic platforms. I am very interested in research that bridges the gap between theoretical AI methods and their deployment in physical environments. I believe that embodied AI presents an exciting opportunity to investigate how perception, learning, reasoning, and action can be integrated to create autonomous systems capable of operating reliably in dynamic and complex environments.

I am highly motivated to pursue a PhD because I want to develop as an independent researcher and contribute meaningful solutions to challenging problems in artificial intelligence and robotics. I am particularly interested in publishing high-quality research and collaborating with researchers from different backgrounds. I am also willing and enthusiastic to strengthen my expertise in robotics-specific

technologies, including ROS/ROS 2, robotic manipulation, robot perception, simulation, and real-world robotic platforms.

I believe my background in Computer Science, combined with my experience in machine learning and AI and my strong motivation to pursue robotics research, would allow me to make a meaningful contribution to the BRAINS Lab. I would be grateful for the opportunity to discuss how my background and research interests could align with your ongoing projects.

Thank you for considering my application. I have attached my detailed CV for your review and would welcome the opportunity to discuss the PhD position and potential research directions.

Sincerely,

Priya Ajay Deshmukh

M.S. in Computer Science

United States